The Power of Noise: Redefining Retrieval for RAG Systems

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Master AI — Information Retrieval 1

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Motivation: When LLMs Know Too Little

Large Language Models are impressive — but limited:

- Knowledge is **frozen** at training time
- Prone to **hallucinations** on factual questions
- Struggle with **domain-specific** or rapidly changing knowledge

Asking a 2023 LLM about last week's events → unreliable answers

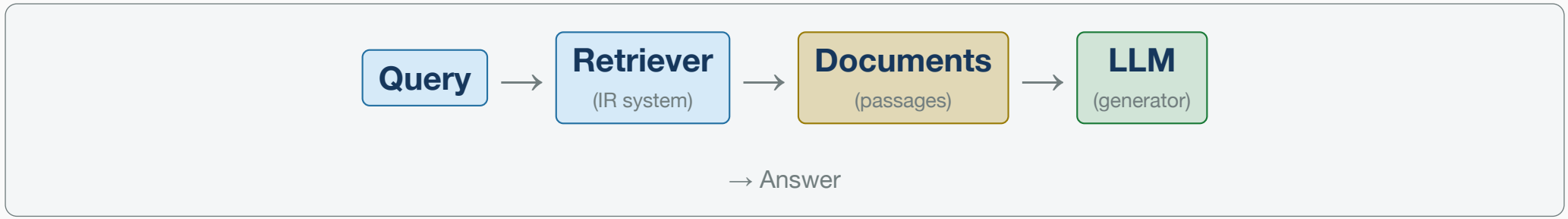
Retrieval-Augmented Generation (RAG) as the fix:

- Augment the prompt with **retrieved passages** at inference time
- Grounds the LLM in real, up-to-date evidence
- Standard in enterprise & search applications (Bing, ChatGPT, etc.)

But: the retriever's role itself has received **surprisingly little** systematic study.

→ This paper asks: **what should a RAG retriever actually retrieve?**

How RAG Works — and What We Don't Know



Common assumption:

- Feed top-k highest-scoring retrieved docs to the LLM
- More semantically relevant = better
- Number of docs limited only by context window

Contributions: first comprehensive study of doc type in RAG; new retrieval heuristics; all code & data released.

This paper's research question:

> *"What characteristics are desirable in a retriever to optimize prompt construction for RAG systems in order to increase the LLM effectiveness?"*
>
> Three dimensions studied: **type**, **position**, and **number** of documents.

A Taxonomy of Retrieved Documents

The paper formalises four document types a retriever can return:

★ Gold — the **ground-truth passage** containing the answer.

● Relevant — other passages that **also contain the answer** (not directly tested).

▲ Distracting — semantically close to the query but **no answer** (top-k but wrong).

■ Random/Noise — **unrelated** to the query; pure noise.

Prompt structure notation:

I	Task instruction
★	Gold document
●	Relevant documents
▲	Distracting documents
■	Random documents
Q	The query

Example: `[I, ▲, ★, Q]` means: instruction, then distracting docs, then gold, then query.

The **position** of each document relative to Q will matter greatly.

Experimental Setup

📄 Dataset

NQ-open (Natural Questions)

- 21M Wikipedia passages (100 words each)
- 2,889 test queries
- Answers ≤ 5 tokens
- Open-domain Q&A

🤖 Generators

- **Llama2-7b** (main model)
- Phi-2
- Falcon-7b
- Mosaic MPT-7b

All open-source, 4-bit quantised
Greedy decoding, max 15 tokens

🔍 Retrievers

- **Contriever** (dense, bi-encoder)
- **BM25** (sparse, term-based)

Dense vs. sparse comparison

Metric: Accuracy — at least one reference answer must appear in the generated response

Significance: Wilcoxon signed-rank test ($p < 0.01$) throughout

Finding 1: Position of the Gold Document Matters

Three positions tested for the gold document ⋆ **Gold** relative to the query **Q**:

Near: `[I, ▲, ⋆, Q]`
Gold adjacent to query

Far: `[I, ⋆, ▲, Q]`
Gold at start of prompt

Mid: `[I, ▲, ⋆, ▲, Q]`
Gold buried in the middle

Near > Far > Mid. The middle of the context is hardest to attend to — worse than the start of the prompt.

Intuition from attention patterns:

- Attention scores are **highest near the query** — and also relatively high at the very start of the prompt
- The **middle** of the context gets the least attention: a U-shaped curve
- This is exactly the **"Lost in the Middle"** effect (Liu et al., 2023)

Practical implication: place relevant docs **immediately before** the query. Start of prompt is second best — never bury them in the middle.

Consistent across all tested models (Llama2, Phi-2, Falcon, MPT).

Finding 2: Top-Ranked Docs That Miss the Answer *Hurt*

Standard assumption in RAG:

- Retrieve top-k documents by similarity score
- Semantically close → helpful context

What the paper shows instead:

Distracting documents ▲ **Distracting** (high retrieval score, **no answer**) **decrease** LLM accuracy vs.
using only the gold document.

Example from the paper:

Query: *"What color was Napoléon's horse?"*

A doc about Joséphine de Beauharnais' horse scores high in the retriever (same historical context), but **does not contain the answer**.

Baseline comparison (Llama2, Contriever, Near):

Only gold	`[I, ⋆, Q]`	**0.564**
Gold + 1 distracting	`[I, ▲, ⋆, Q]`	↓ **0.428**
Gold + 2 distracting	`[I, ▲▲, ⋆, Q]`	↓↓ **0.397**

The **retriever's ranking score is not a reliable signal** for downstream LLM usefulness. High BM25/dense score ≠ helpful in the prompt.

The paper's attention analysis shows the LLM **disproportionately attends** to distracting docs over the gold doc, explaining the wrong answers.

Confirmed with ADORE (hard-negative retriever) — same degradation.

Finding 3: Random Noise *Improves* Accuracy — Oracle Setting

Oracle setting: the gold document is always included. Random Wikipedia passages are added around it.

Prompt: `[I, ▇▇▇, ⋆, Q]` — noise + gold near Q

Results (MPT-7b, Near):

Only gold `[I, ⋆, Q]`	0.215
1. 4 random docs (best)	**0.293**

+36% accuracy improvement from adding documents that contain no answer at all!

Not consistent across all models:

- **MPT:** noise helps in *all* positions (Near, Far, Mid)
- **Llama2 & Phi-2:** noise helps only in the **Near** setting; hurts in Far/Mid
- **Falcon:** noise does *not* improve accuracy in this oracle setting

The benefit of noise is **model-dependent** in the oracle setting — but Near positioning + noise is the most robust combination.

Best noise position: far from query; gold near query. `[I, ▇▇▇, ⋆, Q]` outperforms `[I, ⋆, ▇▇▇, Q]`.

Finding 4: Noise Helps in a Realistic RAG Setting Too

Realistic setting: retrieve with the IR system, then pad remaining context with random noise.

Prompt: `[I, ■ ■ ■, ▲/●, Q]`

Accuracy (Llama2-7b, NQ-open test set):

4 retrieved only (Contriever)	0.187
1. fill with noise (Contriever)	**0.253** (+35%)
4 retrieved only (BM25)	0.203
1. fill with noise (BM25)	**0.293** (+44%)

Noise boost **carries over** to the realistic setting — and holds for different noise sources (Wikipedia, Reddit, random words).

The retriever trade-off:

Few retrieved (3–5)	mostly relevant → high signal
Many retrieved (8+)	distractors creep in → accuracy drops

Falcon: did *not* improve with noise in the oracle setting, but **does improve** in this realistic setting — suggesting the benefit is more general than the oracle results imply.

Sweet spot: retrieve **3–5 docs**, pad context with noise. More docs → more distractors.

Why Does Noise Help? The Attention Entropy Hypothesis

Attention entropy experiment:

Compare `[I, ⋆, Q]` vs `[I, ■■, ⋆, Q]`:

Attention entropy **increases ≈ 3×** when random documents are added.

What is "entropy collapse"?

Prior work (Attanasio et al., 2022; Hoffmann et al., 2023) shows that:

- Pathologically **low** attention entropy → degenerate, repetitive outputs
- Also linked to worse factual grounding

Hypothesis: random documents act as "attention anchors," preventing the model from collapsing onto a narrow region of the context. Higher entropy = more distributed, more robust attention → better answers.

This is an **open hypothesis**, not a proven mechanism. The paper shows **correlation** (noise → higher entropy → better accuracy) but not causation. Explicitly flagged as future work.

Practical Heuristics from the Paper

Proposed retrieval strategy:

- Retrieve **3–5 documents** with your IR system
- Place them **near the query** in the prompt
- Fill the remaining context with **random documents**
- Put random filler **far from the query**

Result: `[I, ■ ■ ■, ▲/●, Q]`

Why not retrieve more?

More retrieved docs → increasing chance of **distracting** content → accuracy degrades.

The sweet spot for the studied setup: $k = 3$ **to** 5.

Key insight for the field:

Retrieval quality for RAG is **not** the same as classic IR relevance ranking. A retriever optimised purely for top-k relevance may **hurt** the downstream LLM. New retrieval objectives are needed.

Connecting to the Course

IR concepts at the core of this paper:

Dense retrieval (DPR / Contriever)

Bi-encoder models, FAISS ANN search — covered in lectures on neural IR

Sparse retrieval (BM25)

TF-IDF weighting, probabilistic relevance framework — core course material

Top-k retrieval & recall

Standard evaluation metric; paper questions whether it predicts downstream usefulness

Retrieval meets generation (RAG)

Discussed as part of modern neural IR systems in class

What this paper adds beyond the course:

> Classic IR optimises for **document relevance to query**. RAG requires optimising for **LLM generation quality given the prompt**. These are **not the same objective**.

Related work from the broader literature:

- "Lost in the Middle" (Liu et al., 2023) — positional bias in LLMs
- DPR (Karpukhin et al., 2020) — dense passage retrieval for OpenQA
- Entropy collapse (Hoffmann et al., 2023) — attention stability in transformers

Conclusions

Three key findings — remember these:

- **Position matters.** Place relevant docs **near the query** — the LLM cannot attend well to distant context.
- **Distracting docs hurt.** Non-relevant top-k docs **reduce** accuracy.
- **Noise helps.** Random docs **improve** accuracy by up to 35–44%.

Published at SIGIR 2024. Code & data released at
`github.com/florin-git/The-Power-of-Noise`
129 citations as of Feb 2026.

Open questions & future work:

- **Why** does noise help? The entropy hypothesis is suggestive but not proven.
- Which properties make a document useful as filler?
- Can we **train** a retriever with an objective aligned to LLM accuracy, not just relevance?
- Do these findings hold for larger LLMs (13B, 70B)?
- How do results change with instruction-tuned / RLHF models?